

NOTCH1 (human, P46531) — Axon Guidance (GO:0007411) Function-Assignment Review

Focus: function_assignment · function-hypothesis-go-0007411 **Source:** genes/human/NOTCH1/NOTCH1-ai-review.yaml **Iteration:** 3 of 3 (final)

Executive Judgment

Verdict: Partially supported / weakly supported — retain only as a non-core, phylogenetically-inferred BP annotation with an explicit caveat; do NOT treat as a direct, experimentally-established human function.

The seed hypothesis is correct on two points and unproven on the decisive one:

1. **A genuine, guidance-specific Notch mechanism exists** — but it has been demonstrated only in *Drosophila*. Notch controls axon growth and guidance through a **noncanonical, Su(H)-independent pathway** using the Abl tyrosine kinase, the adaptor Disabled, the GEF Trio, and Rac GTPase

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This is mechanistically distinct from SLIT/ROBO and from canonical nuclear Notch, exactly as the hypothesis states, and it is separable from neuronal fate.

2. **The "questionable cross-family annotation source" is real and confirmed.** The human NOTCH1 → GO:0007411 record is an **IBA/PAINT phylogenetic inference** (ECO:0000318, GO_REF:0000033, GO_Central), not a human experiment. Its withFrom support set mixes legitimate Notch orthologs (fly Notch FBgn0264089; mouse Notch MGI:1315202/3/5) **with human SLIT1/SLIT2/SLIT3** (O75093/O75094/O94813) — a distinct SLIT/ROBO ligand family. So SLIT cross-family contamination materially contributed to the propagation.

3. **The decisive gap:** there is **no direct human/vertebrate axon-pathfinding evidence**. The only documented mammalian NOTCH1 neuronal phenotype is **inhibition of neurite outgrowth / neurite morphology** via canonical CBF1 signaling

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— a length/morphology effect the hypothesis explicitly asks to *exclude* from guidance evidence. Mammalian noncanonical Notch does exist in neurons but has been mapped to a different output (synaptic-vesicle protein expression,

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A computed STRING v12 check found **no curated interaction** between human NOTCH1 and the Abl/Trio/Rac module (the NOTCH1–ABL1 edge is text-mining-driven: tscore 0.596, escore 0.113, dscore 0).

Bottom line for the curator: The hypothesis's principled stance — that a bad annotation source and the absence of a human assay do not by themselves refute a conserved function — is fair. But "not refuted" is not "supported." The affirmative evidence is entirely *Drosophila*, the human annotation is inference contaminated by a paralogous family, and the accessible human data point *away* from directional guidance and toward neurite morphology/fate. The term should be retained at most as a **non-core, inferred** annotation with a caveat, or generalized; it should **not** be presented as a demonstrated core human function.

Evidence Matrix

Citation	Evidence type	Direction	Claim tested	Key finding	Context
QuickGO / GO_Central (GO_REF:0000033)	Computational / database (IBA)	Qualifies	Is the human annotation experiment- based?	IBA phylogenetic annotation; withFrom mixes fly/mouse Notch with human SLIT1/2/3	Human P46531, PAINT
P 18062953	Direct assay + mutant phenotype	Supports (fly)	Does Notch perform guidance via a guidance- specific step?	Notch directs axon growth/ guidance via Su(H)- independent Abl/Disabled/ Trio pathway; separable from cell fate	<i>Drosophila</i> CNS/motor axons
P 21246649	Mutant phenotype / genetic	Supports (fly)	What is the guidance effector step?	Rac (via Trio GEF1) is the crucial Rho GTPase in noncanonical Notch/Abl guidance	<i>Drosophila</i> motor neurons
P 29343637	Direct assay	Supports (fly)	Receptor- proximal requirements?	Tyrosine phosphorylation + proteolytic cleavage of Notch required for noncanonical Notch/Abl axon patterning	<i>Drosophila</i>
P 10465425	Direct assay (overexpression)	Competing / qualifies	What does mammalian Notch1 do in neurons?	Notch1 inhibits neurite outgrowth / morphology via CBF1	Mouse primary cortical/

Citation	Evidence type	Direction	Claim tested	Key finding	Context
				(canonical); blocked by Numb	hippocampal neurons
P 27040987	Direct assay / genetic	Qualifies	Does mammalian noncanonical Notch exist, and its output?	γ -secretase/CBF1- independent Notch controls synaptic-vesicle protein expression	Mouse postmitotic neurons
STRING v12 (computed this run)	Computational / interaction	Qualifies / refutes- coupling	Does human NOTCH1 couple to Abl/ Trio/Rac?	No ABL1/TRIO/ DAB1/RAC1 in top-80 partners; NOTCH1–ABL1 edge text-mining- driven (t=0.596, e=0.113, d=0)	Human 9606

Provenance artifacts: `NOTCH1_G00007411_evidence_matrix.csv` (this run); QuickGO and STRING API calls executed live in iterations 1–2.

GO Curation Implications (leads — require curator verification)

GO ID	Term	Aspect	Current evidence	Recommended action (lead)	Rationale
GO:0007411	axon guidance	BP	IBA (ECO:0000318)	Retain as non-core inferred, add caveat — OR generalize	No direct human/vertebrate pathfinding evidence; support set SLIT-contaminated; conserved fly mechanism unverified in human

- The evidence supports a **BP** term at most, not an MF. Do **not** downgrade to "protein binding."
- **Do not present as a core function.** If the review pipeline distinguishes core vs. non-core, mark axon guidance **non-core** for human NOTCH1.
- If a curator wants to preserve the conserved biology without over-claiming the human protein, an acceptable lead is to keep GO:0007411 **only** with the IBA evidence code and a note that the phylogenetic support is partly cross-family (SLIT), OR generalize to a less specific neuronal-development term that the mammalian data actually support (e.g., regulation of neuron differentiation / regulation of neuron projection development — curator to verify exact ID).
- **Flag the PAINT annotation to GO_Central:** the `withFrom` inclusion of SLIT1/2/3 in a Notch-family propagation is a likely tree/family-grouping artifact worth reporting.

Mechanistic Scope

- **Immediate molecular step being tested (fly):** transmembrane Notch receptor, after tyrosine phosphorylation and cleavage, physically associates with Disabled and Trio and activates Rac to remodel the growth-cone cytoskeleton — a *bona fide* guidance-relevant step, independent of Su(H)/CBF1 transcription

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- **Downstream / excluded phenotypes:** neuronal fate specification (canonical Notch), neurite outgrowth length and branching (mammalian

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survival, and disease phenotypes. These are explicitly outside the guidance claim and, for human, are the *only* things actually observed.

- **Directionality distinction:** "axon guidance" (GO:0007411) means directional pathfinding/patterning. Mammalian NOTCH1 data show a scalar effect on neurite length/morphology, not steering — an important curation distinction.

Conflicts and Alternatives

1. **Cross-family (paralog/family) confusion:** SLIT1/2/3 in the IBA withFrom set means part of the "axon guidance" signal came from the true SLIT/ROBO guidance ligands, not from Notch biology — classic annotation carry-over.
 2. **Organism-specific difference:** The guidance mechanism is demonstrated in *Drosophila*; the mammalian ortholog's documented neuronal action (neurite inhibition via CBF1) is a *different* pathway output.
 3. **Canonical vs. noncanonical:** The human annotation does not specify the noncanonical mechanism; a naive reader could conflate it with canonical Notch, which is not what the fly guidance work shows.
 4. **Interaction-data gap:** No curated human NOTCH1–ABL1/TRIO/DAB1/RAC1 interaction; the apparent link is literature co-mention, not experiment.
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Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
No human/vertebrate axon-pathfinding assay for NOTCH1	PubMed (multiple queries) — none found	Decides whether the term is a demonstrated human function	In vivo/vitro growth-cone turning or commissural/RGC pathfinding assay with NOTCH1 perturbation
Human NOTCH1↔Abl/Trio/Rac coupling	STRING v12 (computed) — no curated edge	Tests whether the conserved mechanism operates in human	Co-IP/proximity labeling of NOTCH1 with ABL1/TRIO/DAB1 in human neurons
Conservation of the fly Notch phospho-tyrosine/cleavage guidance determinants in human NOTCH1	Not directly aligned this run (time-limited)	Sequence conservation would strengthen "could perform the step"	Alignment of the Drosophila Notch ICD tyrosine sites/Dab-binding region to P46531
PANTHER family PTN002911625 composition	Inferred from withFrom (contains SLIT + Notch)	Confirms the cross-family grouping causing over-propagation	Direct inspection of the PANTHER tree/family membership

Discriminating Tests

1. **Growth-cone turning / directional pathfinding assay** with human NOTCH1 gain/loss (and a Su(H)/RBPJ-uncoupled allele) to separate guidance from neurite length and fate.
2. **Interaction mapping** (co-IP, BioID/APEX) of human NOTCH1 with ABL1, TRIO, DAB1, RAC1 in neurons to test whether the noncanonical module is engaged.
3. **Structure-function:** express a human NOTCH1 carrying the Drosophila Dab-binding/tyrosine determinants vs. wild type; test rescue of a guidance phenotype.
4. **Vertebrate in vivo:** zebrafish/mouse notch1 conditional knockdown in a defined tract (e.g., commissural or motor axons) scored for pathfinding errors distinct from proliferation/fate.

5. **Sequence/orthology:** align P46531 ICD to Drosophila Notch guidance determinants; report conservation conservatively.

Curation Leads (require curator verification)

- **Action lead:** Change GO:0007411 from an unqualified functional claim to an **IBA, non-core, caveated** annotation — or generalize to a neuronal-development term the mammalian data support. Do **not** record axon guidance as a demonstrated core human function.
 - **Reference leads with exact snippets to verify:**
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— "*Notch directs axon growth and guidance in Drosophila via a "non-canonical", i.e. non-Su(H)-mediated, signaling pathway, characterized by association with the adaptor protein, Disabled, and Trio, an accessory factor of the Abl tyrosine kinase*" (establishes fly guidance mechanism).
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— "*it is the crucial Rho GTPase in "noncanonical" Notch/Abl signaling*" (Rac effector step).
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— "*Transfection at two days in vitro with full length Notch1 inhibited neurite outgrowth*" (mammalian effect is neurite length, not guidance).
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— "*a novel Notch signaling is involved in expression of synaptic proteins in postmitotic neurons*" (mammalian noncanonical Notch output ≠ guidance).
 - **Database lead:** Report to GO_Central that the NOTCH1 GO:0007411 PAINT

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 set includes human SLIT1/2/3, indicating cross-family over-propagation.
 - **Suggested curator questions:** (a) Is there any vertebrate in vivo axon-pathfinding phenotype for notch1 not attributable to fate/proliferation? (b) Does the review pipeline want to keep conserved-but-unttested functions as IBA-only with caveats?
 - **Suggested experiments:** items 1–3 under Discriminating Tests.
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Limitations

- Three-iteration budget; PubMed queries intermittently rate-limited and some returned no hits (may under-sample vertebrate literature).
 - Sequence-level conservation of the specific fly guidance determinants was not aligned in this run (flagged as a gap).
 - STRING absence of an edge is suggestive, not proof, of no interaction.
 - Local repository `*-bioinformatics` analyses were intentionally withheld and not consulted.
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